Question	Answer	Marks	Partial Marks
7(c)(ii)	92	**2**	**M1** for 150 −12 oe seen If 0 scored, **SC1** for answer 8[%]
8(a)	$\frac{1}{2}$ or 0.5 oe	**2**	**M1** for $10-3=11p+3p$ oe or better
8(b)	$[m=]\frac{2k}{c^2-g}$ oe final answer	**3**	**M1** for correctly isolating *m* terms **M1** for correctly factorising **M1** for dividing by a bracket with two terms to the final answer Maximum mark **M2** if final answer incorrect
8(c)	0 4.5 oe	**5**	**B4** for $2x^2-9x$ [= 0] or $9x-2x^2$ [= 0] or better OR **M2** for $(2x+3)+4(x-3)=(x-3)(2x+3)$ or better or **M1** for $(2x+3)+4(x-3)$ seen oe or common denominator $(x-3)(2x+3)$ oe **B1** for $2x^2-6x+3x-9$ or better seen
8(d)	$y^2-10y+21[=0]$ or $x^2-4x-12[=0]$	**M2**	**M1** for $y^2+5(12-2y)=39$ oe or $5x+\frac{(12-x)^2}{2^2}=39$ seen oe
	$(y-3)(y-7)$ [= 0] or $(x+2)(x-6)$ [= 0]	**M1**	or for correct factors for *their* 3– term quadratic equation or for correct substitution into quadratic formula or correctly completing the square for *their* 3– term quadratic equation
	$x=-2$ $y=7$ $x=6$ $y=3$	**B2**	**B1** for $x=-2$, $x=6$ or for $y=7$, $y=3$ or for one correct pair of *x* and *y* values
8(e)	$2x^3+x^2-54x+72$ final answer	**3**	**B2** correct expansion of three brackets unsimplified or for final answer of correct form with 3 out of 4 terms correct or **B1** correct expansion of two brackets with at least three terms out of four correct
9(a)	*PMR* = *MSR* = right angle[s] or 90°	**B1**	
	PRM = *MRS* same angle	**B1**	
	AAA oe OR *MPR* = *SMR* 3rd angle of triangle	**B1**	**Dep** on B1B1 and no errors seen

Question	Answer	Marks	Partial Marks
9(b)(i)	5.5	**2**	**M1** for $\frac{x}{4.5}=\frac{9.9}{8.1}$ oe
9(b)(ii)	16.7 or 16.73 to 16.74	**2**	**M1** for $25\times\left(\frac{8.1}{9.9}\right)^2$ oe or $25\times\left(\frac{4.5}{\textit{their } 5.5}\right)^2$ oe
10(a)	1, 2, 3	**2**	**M1** for $15-8>5n-3n$ oe If 0 scored, **B1** for 2 correct answers and no others or 3 correct answers with one extra value
10(b)(i)	$10y + 8x \leqslant 80$ oe final answer $x > 4$ oe final answer $2y > x - 4$ oe final answer	**3**	**B1** for each If 0 scored, **SC1** for $10y + 8x < 80$ oe final answer and $x \geqslant 4$ oe final answer and $2y \geqslant x - 4$ oe final answer
10(b)(ii)	23 final answer	**2**	**M1** for 7 and 2 selected soi
11(a)(i)	4.455 to 4.456… [= 4.46]	**2**	**M1** for $[r =]\ \frac{28}{2\pi}$ oe
11(a)(ii)	1250 or 1247 to 1249.9…	**2**	**M1** for $20\times\pi\times4.46^2$ oe
11(a)(iii)	66[.0] or 65.95 to 66.02	**3**	**M2** for [tan] = $\frac{20}{2\times4.46}$ oe or **B1** for identifying angle *ANB* on cylinder **not on rectangle**
11(b)	11.8 or 11.82 to 11.83	**5**	**M2** for $[r =]\ \sqrt[3]{\frac{310\times3}{2\pi}}$ oe or $[h =]\ \sqrt[3]{\frac{310\times3\times4}{\pi}}$ oe or **M1** for $310=\frac{1}{3}\pi\times r^2\times2r$ or $310=\frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h$ **M2** for $\sqrt{(\textit{their } r)^2+(2\times\textit{their } r)^2}$ oe or **M1** for $[l^2 =](\textit{their } r)^2+(2\times\textit{their } r)^2$ oe